

Lab Instrument Survey

Detail about the equipment, the PCs attached to it and the site around it. It is a form rather than a meeting because most of this is easier to read off the equipment than to recall.

Laboratory

Completed by

Role

Date

Issued with

Answers

Return to

Assay Lab Requirements, section 8

questions Q-01 to Q-04 and Q-16 to Q-19

Before you start

Please fill in one block per instrument in section 1 — there are five, and blank ones can be ignored or the page copied if there are more. Two people will probably be needed: someone from the laboratory for the instrument itself, and someone from IT for the PC attached to it.

If a field cannot be answered, leave it blank rather than guessing. A blank tells us to come and look; a guess sends us down the wrong road.

THE ONE THING WORTH MORE THAN THE REST OF THIS FORM

An example output file from each instrument — whatever it produces after a run, exactly as it produces it. One real file answers questions this whole form only approximates, and it is the difference between estimating instrument integration and guessing at it.

Instruments

One block per balance, spectrometer or analyser, including any that are old, rarely used, or currently out of service. Instruments nobody mentions are the ones that derail an integration.

01 Balance

THE INSTRUMENT

Instrument type

☐ Balance ☐ AAS ☐ LECO ☐ Other

Make

Model

Serial or asset number

Location in the laboratory

Approximate year acquired

THE PC ATTACHED TO IT

PC name or asset tag

Windows version

On the site network?

☐ Yes ☐ No ☐ Not sure

Who holds administrator rights on this PC

HOW ITS DATA COMES OUT TODAY

Output method – tick every one that applies

☐ Serial cable (RS-232) ☐ USB ☐ Writes a file to a folder ☐ Prints only

☐ Read from the screen ☐ Writes to a database

If it writes a file: format and folder location

Spare serial or USB port free?

☐ Yes ☐ No ☐ Not sure

Example output file attached to this form?

☐ Yes ☐ Not available

USE AND UPKEEP

Who operates it, and on which shifts

Readings produced per day

Calibrated or serviced by whom, how often

Still supported by the vendor?

Anything unusual we should know about this instrument

02 Balance

THE INSTRUMENT

Instrument type

☐ Balance ☐ AAS ☐ LECO ☐ Other

Make

Model

Serial or asset number

Location in the laboratory

Approximate year acquired

THE PC ATTACHED TO IT

PC name or asset tag

Windows version

On the site network?

☐ Yes ☐ No ☐ Not sure

Who holds administrator rights on this PC

HOW ITS DATA COMES OUT TODAY

Output method – tick every one that applies

☐ Serial cable (RS-232) ☐ USB ☐ Writes a file to a folder ☐ Prints only

☐ Read from the screen ☐ Writes to a database

If it writes a file: format and folder location

Spare serial or USB port free?

☐ Yes ☐ No ☐ Not sure

Example output file attached to this form?

☐ Yes ☐ Not available

USE AND UPKEEP

Who operates it, and on which shifts

Readings produced per day

Calibrated or serviced by whom, how often

Still supported by the vendor?

Anything unusual we should know about this instrument

03 AAS

THE INSTRUMENT

Instrument type

☐ Balance ☐ AAS ☐ LECO ☐ Other

Make

Model

Serial or asset number

Location in the laboratory

Approximate year acquired

THE PC ATTACHED TO IT

PC name or asset tag

Windows version

On the site network?

☐ Yes ☐ No ☐ Not sure

Who holds administrator rights on this PC

HOW ITS DATA COMES OUT TODAY

Output method – tick every one that applies

☐ Serial cable (RS-232) ☐ USB ☐ Writes a file to a folder ☐ Prints only

☐ Read from the screen ☐ Writes to a database

If it writes a file: format and folder location

Spare serial or USB port free?

☐ Yes ☐ No ☐ Not sure

Example output file attached to this form?

☐ Yes ☐ Not available

USE AND UPKEEP

Who operates it, and on which shifts

Readings produced per day

Calibrated or serviced by whom, how often

Still supported by the vendor?

Anything unusual we should know about this instrument

04 LECO

THE INSTRUMENT

Instrument type

☐ Balance ☐ AAS ☐ LECO ☐ Other

Make

Model

Serial or asset number

Location in the laboratory

Approximate year acquired

THE PC ATTACHED TO IT

PC name or asset tag

Windows version

On the site network?

☐ Yes ☐ No ☐ Not sure

Who holds administrator rights on this PC

HOW ITS DATA COMES OUT TODAY

Output method – tick every one that applies

☐ Serial cable (RS-232) ☐ USB ☐ Writes a file to a folder ☐ Prints only

☐ Read from the screen ☐ Writes to a database

If it writes a file: format and folder location

Spare serial or USB port free?

☐ Yes ☐ No ☐ Not sure

Example output file attached to this form?

☐ Yes ☐ Not available

USE AND UPKEEP

Who operates it, and on which shifts

Readings produced per day

Calibrated or serviced by whom, how often

Still supported by the vendor?

Anything unusual we should know about this instrument

05 Spare — any other instrument

THE INSTRUMENT

Instrument type

☐ Balance ☐ AAS ☐ LECO ☐ Other

Make

Model

Serial or asset number

Location in the laboratory

Approximate year acquired

THE PC ATTACHED TO IT

PC name or asset tag

Windows version

On the site network?

☐ Yes ☐ No ☐ Not sure

Who holds administrator rights on this PC

HOW ITS DATA COMES OUT TODAY

Output method – tick every one that applies

☐ Serial cable (RS-232) ☐ USB ☐ Writes a file to a folder ☐ Prints only

☐ Read from the screen ☐ Writes to a database

If it writes a file: format and folder location

Spare serial or USB port free?

☐ Yes ☐ No ☐ Not sure

Example output file attached to this form?

☐ Yes ☐ Not available

USE AND UPKEEP

Who operates it, and on which shifts

Readings produced per day

Calibrated or serviced by whom, how often

Still supported by the vendor?

Anything unusual we should know about this instrument

02

The site

Where the system would live, and what it can rely on.

SITE Environment, network and support

Is there a server room or comms cabinet in or near the laboratory? Where?

Existing servers in the laboratory – make, operating system, what they do

The server that data is copied to at the end of the day: what is it, and what is stored on it?

Mains power

☐ Stable ☐ Frequent outages ☐ Generator present ☐ UPS present

Internet connection type

Typical reliability

Typical speed if known

Wifi coverage

☐ Laboratory ☐ Sample reception ☐ Prep area ☐ Field camp ☐ None

Who provides IT support on site, and during which hours

How are systems backed up today?

Any policy restricting hosted systems, or data leaving the country?

☐ Yes – describe below ☐ No ☐ Not sure

If yes, what does the policy require?

03

Volumes and people

Used for sizing the system and the hardware. Approximate figures are fine; the peak matters more than the average.

SCALE Throughput, staffing and retention

Samples per day – typical	Samples per day – peak	Busiest month of the year
_____	_____	_____
Batches fired per day – typical	Batches fired per day – peak	Standard batch size
_____	_____	_____
Shift pattern worked	Number of laboratory staff	Number who would use the system
_____	_____	_____
Staff by role – reception, preparation, assay, quality control, management		

Accredited to ISO/IEC 17025, seeking accreditation, or neither?	How long must records be kept?	
_____	_____	

04

Documents to send with this form

Tick what you are able to provide. Anything commercially sensitive can be redacted – we need the structure of these documents, not their contents.

- ☐ **A completed sample submission or dispatch sheet**
One that actually came in with samples, filled in, rather than a blank template.
- ☐ **A fire assay batch sheet**
As used at the bench, showing how positions are laid out and where controls sit.
- ☐ **A results report as issued, in Excel**
The working file, not a printout.
- ☐ **A results report as issued, as a PDF on letterhead**
So the report template can be matched exactly.
- ☐ **A doré or bullion bar certificate**
- ☐ **Certificates for the reference materials currently in use**
Certified values and standard deviations are needed to configure the control rules.
- ☐ **Standard operating procedures for preparation and fire assay**
- ☐ **One example output file from each instrument**
The single most useful item on this list. A raw file tells us in minutes what a meeting cannot.
- ☐ **Examples of the current sample numbering**

Including any prefixes or site codes that must be preserved.

Please return the completed form together with the example instrument output files. Questions Q-01 to Q-03 in the requirements document depend on section 1 of this survey; until it is returned, integration effort cannot be estimated.